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1 CHAPTER TWO "Text Analysis"

1.1 Doubt

1

The novel *Orlando: A Biography* (1928), by Virginia Woolf, famously opens with an assertive gender designation followed by an immediate qualification:

He—for there could be no doubt of his sex, though the fashion of the time did something to disguise it— (italics mine, 11)

2

In order to perform quantitative text analysis on this text, the standard tasks of "pre-processing" or "cleaning" would evacuate this sentence of its gender qualification. Cleaning the first sentence not only would strip it of its pronouns, including the gender assertion in the first word, "He," it would also cut the em dash immediately following "He," signaling the entrance of the narrator and his conspicuous certitude, "—for there could be no doubt of his sex." After cleaning, the first sentence would appear as the following list of words:

3

‘could’, ‘doubt’, ‘sex’, ‘though’, ‘fashion’, ‘time’, ‘something’,
‘disguise’

Text analysis, a process which involves calculating and visualizing textual patterns, requires transforming the source text into computable units. Cleaning the text strips capitalized words, punctuation, “stop words” (such as articles and prepositions), and inflections in word endings, all of which are deemed to be semantically minor, so that they don’t affect the analysis

of more substantial features like nouns, verbs, adverbs, and adjectives. Text analysis requires removing such "minor" elements of language in order to find connections among more apparently substantial terms.

4

However, the word "He," and the em dash following and accentuating that "He," is hardly minor in a text that interrogates gender differences and whose protagonist experiences their own shift in gender. Indeed, as Pamela Caughie puts it, the pronoun and punctuation in the first sentence effectively announces the book's investment in subverting gender norms: "immediately our doubt is aroused... The emphasis on an innocent pronoun makes it suspect" ("Double Discourse" 42).

5

This chapter examines how quantitative text analysis might be used to analyze gender, using Woolf's *Orlando* as a test case. It explores an experimental approach to text analysis that draws from computer programming and gender theory, Judith Butler's theory of gender performativity. This analysis surfaces a mechanism, specifically a technical constraint, common to both text analysis and gender performativity, which is *iteration*. I demonstrate that the logic of repetition that drives iteration can be re-worked into a new text analysis procedure that brings previously suppressed details about gender, and the gender binary, to the surface. The final section of the chapter thus illustrates a text analysis method that re-signifies the meaning of binary gender terms, of "man" and "woman." It works within the constraining structure of iteration in order to destabilize the binary from the inside.

1.2 Falsifiable

1

The history of text analysis as a practice has been traced to many origins.¹ My particular version begins with Franco Moretti, whose seminal book on "distant reading," published as a series of polemical essays in a collection of the same name in 2008, dominated discussions of digital humanities and literary studies I was a graduate student. Distant reading, by Moretti's account, involves counting up textual features to derive textual patterns

¹One rendition traces this lineage to Roberto Busa, an Italian Jesuit priest who authored the first digital concordance, the *Index Thomisticus* (See Ramsay, Stephen, *Writing Machines*. 2011). Another traces this practice to the intersection of literary studies and social science, centering in particular the work of marxist scholar Raymond Williams and feminist scholar Janice Radway (See Ted Underwood, "A Genealogy of Distant Reading").

and make inferences. It is an analytical perspective where "distance," he explains, "is a condition of knowledge" (*Distant Reading* 48).

2

Shortly after the publication of a *New York Times* profile of the scholar, three of his former students came forward to accuse him of sexual crimes. The accusations ranged from unsolicited touching to rape. At the time of the alleged incident, one of his accusers, Kimberly Latta, was a first-year graduate student.

When I was a graduate student at UC Berkeley in 1984-85, my then-professor Franco Moretti sexually stalked, pressured, and raped me. He specifically said to me, 'You American girls say no when you mean yes.'[...] I reported him to the Title IX officer [...] and he threatened to ruin my career if I pressed charges against him. He said he had powerful friends who were attorneys who would ruin my name. I remained silent for all these years because I was in academia" (Kimberly Latta "Op-Ed: What Happened?" 2017).

When the women came forward, Stanford went no further than claiming they would review the cases. There has been no record of an inquiry, investigation, or any repercussions for Moretti (Liu, Knowles, "Harassment, assault allegations against Moretti span three campuses"). He remains an Emeritus Professor at Stanford.

3

At the time of the alleged assault, Moretti was riding his first wave of academic celebrity. Invited to teach at Berkeley, he had just published a collection of essays, *Signs Taken For Wonders: On the Sociology of Literary Forms* (1983), a book which already expressed his polemical style and sowed the roots of his quantitative approach to studying literature. The first essay, "The Soul and the Harpy," takes traditional literary criticism as its target, critiquing the methodology of close reading. According to Moretti, close reading is overly subjective, turning the critic to a kind of "Narcissus", "whose only pleasure lay in contemplating his own reflection ("Soul" 14). He proposes an alternative, "falsifiable criticism," a methodology that pursues answers which are irrefutable. He asserts that,

"The day criticism gives up the battle cry 'it is possible to interpret this element in the following way,' to replace it with the much more prosaic 'the following interpretation is impossible for

such and such a reason,' it will have taken a huge step forward on the road of methodological solidity." ("Soul" 22)

Moretti implicitly aligns literary criticism with methodologies from the social sciences, methods which are traditionally described as "reproducible."² Reproducible methodologies obviate the risk of alternative interpretations by supplying conclusions that are, in Moretti's words, "coherent, univocal, and complete" ("Soul" 21).

4

Twenty years later, falsifiable criticism finds an ideal toolset in quantitative practices. Rather than close attention to individual samples, quantification counts patterns of specific features across wide swaths of text. As we will see below, it simplifies text into workable units, which is the basis for higher levels of computation. These simple and workable units, at large scales, enable ambitious interpretations. As Moretti explains, "If we want to understand the system in its entirety, we must accept losing something. We always pay a price for theoretical knowledge" (*Distant Reading* 49).

5

As a graduate student, I found Moretti's writing both striking and alluring. He is an assertive writer, using mostly short sentences that give his prose a plucky rhythm. Moreover, despite drawing from disparate fields of knowledge, such as evolutionary theory, systems theory, and Marxism, his arguments seem to circle methodically around a drain so that, as long as you're following him, there appears to be no alternative. Paragraphs begin with sentences like,

So, the market selects the canon. (69)

So. As novelistic forms travel through the literary system, their plots are (largely) preserved, while their styles are (partly) lost—and replaced by 'local' ones. (134)

What did the birth of a consumer society mean for the European novel? More novels, less attention. (175)

The major metamorphosis of eighteenth-century titles is simple: in the space of two generations, they become much, much shorter.

²Sarah Cohen-Boulakia, Khalid Belhajjame, Olivier Collin, Jérôme Chopard, Christine Froidevaux, Alban Gaignard, Konrad Hinsén, Pierre Larmande, Yvan Le Bras, Frédéric Lemoine, Fabien Mareuil, Hervé Ménager, Christophe Pradal, Christophe Blanchet, Scientific workflows for computational reproducibility in the life sciences: Status, challenges and opportunities, *Future Generation Computer Systems*, Volume 75, 2017, Pages 284-298, ISSN 0167-739X <https://doi.org/10.1016/j.future.2017.01.012>.

(182)

As the market expands, titles contract; as they do that, they learn to compress meaning; and as they do *that*, they develop special 'signals' to place books in the right market niche. (204)

These sentences, taken from mostly different essays in *Distant Reading*, synthesize arguments and evidence from previous paragraphs. They are more summary of previous points than new argumentation. Yet, their apparent solidity, the lack of a subjective "I" to ground them, is striking. It showcases Moretti's characteristic way of transforming inferences, premises, and hypotheses into declarative statements, stating a claim or hypothesis as an objective or self-evident fact. It suppresses the possibility of alternative views—falsifying them before they have the chance to be heard.³

6

In one of the more famous essays from *Distant Reading*, "Style, Inc.: Reflections on 7000 Titles (British Novels, 1740-1850)" Moretti uses text analysis to study the market for British novels from 1740 - 1850. Here, he explores what titles, which shorten in length over the decades, reveal about the rise and fall of certain genres in the market. He outlines his analytical process:

[F]irst, I describe a major metamorphosis of eighteenth-century titles, and try to explain its causes; next, I suggest how a new type of title that emerged around 1800 may have changed what readers expected of novels; and finally, I make a little attempt at quantitative stylistics, examining some strategies by which titles point to specific genres. Three sections, three pieces in the large puzzle of the literary field. (*Distant Reading* 181-2; emphasis mine)

³The faith in the falsifiable, in the power of supposedly objective processes of quantitative computation, is powerful and contagious. It spreads even to those who believe that it has no place in Literary Studies. A controversial detraction by Nan Z. Da claims there is a "fundamental mismatch between the statistical tools that are used and the objects to which they are applied"—that is, to literary objects (620, 601). Yet, despite her oppositional stance toward quantitative tools, Da has something in common with those who champion them: she *agrees* about the objectivity of their procedures. As Ben Schmidt points out, "Far more than anyone I've seen in any humanities article, she asserts that scientists do something arcane, powerful, and true" (Schmidt, "A computational critique of a computational critique of computational critique."). Despite their vastly different views on the role of quantitative methods for studying literature, Da and Moretti appear to agree that these methods ought to provide results that are, at the very least, reproducible.

Moretti's critical method involves posing hypotheses, assembling and analyzing data, making inferences, and occasionally, reframing the original hypotheses. Yet, he repeatedly understates his interpretive moves at each stage of his analysis: "try to explain," "suggest," "make a little attempt." As Stephen Ramsay points out, "data is presented to us... not as something that is also in need of interpretation" (Ramsay, *Reading Machines* 5). Proffering his insights as a description of reality, Moretti misrepresents the role that inferential thinking plays in his analysis, which contain provocative conclusions, for example, that titles function as a proxy for market forces, as "a code, in the market... where the novel as language meets the novel as commodity" (*Distant Reading* 181).

8

Crucially, as Lauren Klein points out, distant reading can overlook details which are necessary to cultural critique. In a talk titled "Distant Reading After Moretti," Klein explains,

[I]t's not a coincidence that distant reading does not deal well with gender, or with sexuality, or with race. Gender and sexuality and race are precisely the sorts of concepts that have been exposed and interrogated by attending to non-dominant subject positions. And like literary world systems, or "the great unread," the problems associated with these concepts, like sexism or racism, are also problems of scale, but they require an increased attention to, rather than a passing over, of the subject positions that are too easily (if at times unwittingly) occluded when taking a distant view.

Klein's own work tracing these elements proves how careful attention to what is minor or oblique can surface previously suppressed stories. In one project, Klein uses network analysis methods to bring forth the "ghost" of a previously obscured subject, James Hemmings. Hemmings, a slave who worked for Thomas Jefferson until he could negotiate his freedom, appears in brief references in Jefferson's correspondence. By marking up and visualizing these oblique references, Klein "render[s] visible the archival silences implicit in our understanding of chattel slavery today" (665). Her work demonstrates that sometimes, entire lives are deeply deeply distributed text, and require close and careful reading.

10

Indeed, the faith in the falsifiable may be a proxy for avoiding certain kinds of cultural critique. More than avoidant, I want to argue, falsifiable criticism is intellectually coercive. It functions through an amplification and

valorization of what is seemingly objective—in particular, the counting of words. With word counts and their resulting graphs and charts doing the work of analysis, one finally closes down the pesky, subjective interpretations associated with close reading. As Moretti says, "we know how to read texts, now let's learn how *not* to read them" (*Distant Reading* 48).

But more than that, it can also be a way of silencing less prominent or otherwise inconvenient views. As an imposition of a view from the top, it functions like other hierarchies of power, like academic prestige. Like when an older, accomplished male professor gets away with telling a young, female graduate student that "You American girls say No when you mean Yes".

11

Since Moretti, however, there have been many literary scholars that push back against the apparent objectivity of reproducible methods. Ted Underwood, for example, explains that "algorithms are actually bad at being objective and rather good at absorbing human perspectives implicit in the evidence used to train them" ("Machine Learning and Human Perspective" 92). Underwood creates what he calls "perspectival models" of literary data, where models represent different datasets associated with different "perspectives." In one project, Underwood uses a binary classification algorithm to cluster terms that male and female writers tend to associate with either gender.

His results show that both male and female writers tend to agree on terms that describe the male gender, like "pocket," "beard," and "grinned," and the female gender, like "hair," "eyes," and "mother". When writers describe characters of their own gender, however, there is disagreement: writers tend to associate their own gender with terms like "wondered," "mind," and "head," and the opposite gender with terms like "meant," "kissed," "tone." The results, which Underwood presents in a visual chart showing two diagonals crossed in opposition, dramatizes a perspectival shift between the two genders.

12

In this example, the reinscription of the "binary structure" happens not only in the choice of model, a classification (i.e. binary) model which makes predictions on a yes/no or, in this case, a male/female scale. It also happens with the choice in visualizing the results in a way that places male and female genders in opposition. The diagonal reinforces a stereotypical active/passive binary (among other binaries) between men and women and, although based on agreement between the genders and their relationships to themselves, nonetheless imposes an *opposition* between how male and female authors describe each other.

When explaining the diagram, Underwood himself admits that "gender theorists will be frustrated by the binary structure of the diagram... I needed a simple picture, frankly, in order to explain how a quantitative model can be said to represent a perspective" ("Machine Learning and Human Perspective" 98). Beyond this reification of the gender binary, however, Underwood's visualization effectively deploys the binary *as a model* for proving the value of quantitative methods. In order for his experiment to take place, the gender binary is a necessary presumption, a "ground truth," that makes the choice of a classification model appear appropriate. In other words, before he even constructs the diagram, gender roles have been already manufactured into opposition.

14

Posing one gender against another is not, I believe, as provocative as work that explicitly deconstruct these categories (and categorization) in the first place. For example, Laura Mandell, who uses stylometry to study sentimental and anti-Jacobin novels, creates new categories that draw gender into genre. She creates doubly refracted categories like, "men writing as men," "women writing as women," "women writing as men," "men writing as women."⁴

15

Applying text analysis to racial categories, Edwin Roland and Richard Jean So explore whether a machine can correctly identify the race of a novel's author. The point, they are careful to say, is not to test whether the machine is correct, but to examine how the categories themselves are constructed. As they explain, "the machine simply measures the robustness of the social constructedness of these given categories and points out what gives them such vigor" ("Race and Distant Reading" 69). Analyzing a large corpus of novels by white and black authors, So and Roland find that black authors generally display more varied vocabulary than white authors. From this result, they infer that White authorship only coheres *as a category* against the variance of Black authorship. In other words, without Black authorship, White authorship would cease to be visible as a coherent category.

Interestingly, this quantitative exercise points Roland and So toward a peculiarity in the results: that the algorithm wrongly categorizes Black-authored novel, James Baldwin's *Giovanni's Room* (1956), as being written by a White author. This misclassification is due to a single word—"appalled"—whose presence, according to the model, has generally been a signal of White

⁴Mandell, Laura. "Gender and Cultural Analytics:: Finding or Making Stereotypes?" par. 35.

authorship. The term only occurs once in the text, in an early scene where the narrator David describes his strained relationship to his father: "I did not want to be his buddy. I wanted to be his son. What passed between us as masculine candor exhausted and appalled me" (Rpt. in Roland, So 71; emphasis mine). Given the connotation of whiteness in "appalled" and its middle French root, "apalir" (to grow pale), Roland and So explain that this term—which emerged as a result of the computer's mistake—enables their analysis to connect gender to race: "the moment David develops a troubled relationship to normative masculinity [as] also the moment he becomes 'White'" (71).

16

Like Underwood, So and Roland began with the binary form, the binary categories of Black/White. But unlike Underwood, they ended with its fracturing. This fracturing, evident in the computer's misclassification of authorship in *Giovanni's Room*, leads to a reading which coordinates race with gender and sexuality. This "queering the machine's color line," the authors explain, opens up new readings of intersectionality in the novel which "in their current form, our data and model are not robust enough to handle" (72). Rather than reproducible, this method is fracturing and refractive, both reflecting and corrupting the constructed nature of binary structures back at the critic. It brings us closer to the methodology that I outline in the next section, *iteration*.

1.3 Iteration in gender

2

I now turn to Judith Butler's theory of gender performativity,⁵ and to a

⁵Some text analysis projects take the language of performance and qualities of the performative to describe and conceptualize their methodologies. For example, Mandell's project takes the behavioral quality of gender performativity and applies it to genre, asserting that both gender and genre "are... highly imitable," so that "anyone can adopt gendered modes of behavior, just as anyone can write in genres stereotypically labeled M/F" (par. 30). In Computational Literary Studies (CLS), Bode operationalizes the "performative" to describe the analytical act itself, which intervenes and becomes inseparable from the object of analysis. "Performativity," they argue, is "attuned to the coconstitution of computational methods and objects" ("What's the Matter with Computational Literary Studies"). Their research explores how "meaning is not carried by texts but produced in situated interactions between readers, texts and contexts," including computational tools. From the field of Computational Humanities, Kent K. Chang, Anna Ho and David Bamman cite performativity as motivation to computationally "measure subversion" in TV show dialogue, to predict whether samples of a TV show, *The Big Bang Theory*, are queer-coded. Bode, Katherine. "Data beyond representation: From computa-

particular mechanism that drives gender performativity: *iteration*. As I explain below, iteration is an essential and constraining mechanism that binds performativity to its effects. However, as Butler explains, this constraint can be exploited to subvert the norm.

My focus on iteration addresses a common misreading of performativity, that it denotes a series of acts that can be imitated at will. Indeed, Butler's book, *Bodies that Matter: on the Discursive Limits of Sex* (1996), which follows *Gender Trouble*, explicitly aims to clarify how performativity is compulsory, bounded on all sides by social norms in ways that delimit a subject's agency. In *Bodies that Matter*, Butler explains that gender performativity is a pre-determined procedure that *constitutes* subjectivity: "there is no subject who decides on its gender. . . . on the contrary, gender is part of what decides the subject" (ix).

In Butler's theory, iteration is a *perpetual re-creation of norms*, which effectively *produces and re-produces power*. As Butler explains, "performativity must be understood not as a singular or deliberate 'act,' but, rather, as the reiterative and citational practice by which discourse produces the effects that it names" (xii).

For example, according to Butler, it is through performativity that "sexed" bodies emerge. They explain that "'Sex' is not a simple fact or static condition of a body, but a process whereby regulatory norms materialize 'sex' and achieve this materialization through a forcible reiteration of those norms" (xii). What appears as physical or biological category is actually a discursive "sedimentation," formed through continual repetition.

Despite this compulsive repetition, performativity can be subverted, according to Butler. Subversion is possible through manipulating the iterative process. This manipulation brings something something unexpected and extraneous into performativity, while nonetheless staying within the boundary of the permissible.

To theorize what is permissible and what is not, Butler draws from post-structuralist feminist theorist Luce Irigaray's theory of the internal vs the external feminine. Taking philosophical discourse as an example, Irigaray argues that philosophers, like Plato, Aristotle, Descartes, among others, represent the gender binary "on the basis of masculine parameters" (18). Irigaray argues that the binary, constructed along masculine terms, precludes the representation of the the feminine, which is effectively excluded from the binary system. The feminine that exists within the system, the internal feminine,

tional modelling to performative materiality." MLA Convention, Seattle, 10 January 2020. <https://katherinebode.wordpress.com/home/mla-convention-2020/>

is a "domesticated" feminine.

Although this external feminine is unrepresentable, it is the *matter* from which the Male/Female binary comes into intelligibility: "The economy that claims to include the feminine as the subordinate term in a binary opposition of masculine/feminine excludes the feminine, produces the feminine as that which must be excluded for that economy to operate" (10). As it happens, Butler refers to the excluded feminine as a *matrix*. With its etymological associations to "mother," "mater," and the "womb," the matrix is a generative force, "an inscriptional space that... designates the constitutive outside of the Platonic economy; it is what must be excluded for that economy to posture as internally coherent" (12-13).

It is not possible, according to Butler (following Irigaray), to invoke the excluded feminine outside of masculine parameters. However, subversion is possible by breaking open performativity processes, specifically, the process of iteration. For performativity, whereby meaning is signified and re-signified endlessly, is never complete. Butler explains that,

This "being a man" and this "being a woman" are internally unstable affairs. They are always beset by ambivalence precisely because there is a cost in every identification, the loss of some other set of identifications, the forcible approximation of a norm one never chooses, a norm that chooses us. (86)

By this "cost in every identification," Butler means all of the identifications which are effectively rejected when one makes a claim to identity. Gender identity must be, through iteration, stabilized repeatedly through a field of potential exclusions.

8

And it is the material of the excluded which can come back, Butler says, as a "troubling return" (xxx). As Butler explains, every "identification takes place through a repudiation which produces a domain of abjection" (xiii). These repudiated, disavowed identities offer "a field of disruptive possibilities," where terms may draw from rejected meanings to be resignified. Such resignifications "threaten to expose the self-grounding presumptions of the sexed subject" (xiii).

9

Butler offers an example with the term "queer," which is a term that harnesses its own history of repudiation for political purposes. The adoption of this term by Gay and Lesbian activism in the 80s and 90s "resignif[ies] the abjection of homosexuality into defiance and legitimacy" (Butler xxviii).

This turning of abjection into legitimacy shows how one attains some kind of agency even within the terms of subjection.

The compulsion to repeat an injury is not necessarily the compulsion to repeat the injury in the same way or to stay fully within the traumatic orbit of that injury. The force of repetition in language may be the paradoxical condition by which a certain agency—not linked to a fiction of the ego as master of circumstance—is derived from the impossibility of choice. (183)

It is through the repetition of repudiated meanings, which necessarily draw from insulting and/or painful pasts, come to alchemize new significations. If a repudiated meaning, a "troubling return," is repeated enough times, Butler argues, the stability of the original identity will be brought into question. The paradox is that what appears to be a cyclical or even robotic behavior takes on an agentic capacity.

10

Returning to Irigaray, Butler explains that Irigaray achieves a kind of subversion of the masculinist system of representation by "mim[ing] philosophy... and, in the mime, tak[ing] on a language that effectively cannot belong to her" (12). Butler cites a passage where Irigaray "mimes" the style of Plato. By taking on Plato's language, Irigaray "performs a repetition and displacement of the phallic economy" (Butler 18). In an interesting double inhabitation, where Butler mimicks Irigaray mimicking Plato, Butler writes,

I will not be a poor copy in your system, but I will resemble you nevertheless by miming the textual passages through which you construct your system and showing that what cannot enter it is already inside it (as its necessary outside), and I will mime and repeat the gestures of your operation until this emergence of the outside within the system calls into question its systematic closure and its pretension to be self-grounding. (18)

Irigaray undermines the authority of the masculine through repetition, "cit[ing] Plato again and again," in a re-iterative practice (18). This practice has a cumulative effect, which is to "expose precisely what is excluded" from the system, and "to reintroduce the excluded into the system itself" (Butler 18). Through iteration, one displaces the expectation, the norm, to suggest something else. This something else, however, is not meant to replace the norm, but to undermine its pretensions. Butler explains that "This is a taking of his [Plato's] place, not to assume it, but to show that it is *occupiable*...

a kind of overreading which mimes and exposes the speculative excesses in Plato (emphasis original, 11).

11

This iteration between the internal and external (repudiated) feminine is another place where Butler's theory is often misunderstood. For example, one text analysis project, cited above, argues that "subversion" may occur through straight characters that employ queer-coded language. Analyzing the transcripts of a popular TV show, *The Big Bang Theory*, the authors claim that "an undercurrent of emotional intimacy and dependency... challenges the rigid boundaries of what is considered acceptable" (Chang et al., 10). However, the characters in question remain within heterosexual relationships, with their admittedly romantic language being relegated to punchlines. Like some comedies which feature crossdressing (*Some Like It Hot*, *Tootsie*), which Butler describes as "drag as high het entertainment," this show "produce[s] and contain[s] the homosexual excess" in a way that ends up propping up the heterosexual regime (85). Rather than the excluded or repudiated queerness, it is closer to what Irigaray describes as a "domesticated" queerness.

12

Subversion then, works toward displacement through occupation and resemblance. One way that it occurs is through iteration, as Irigaray does with her mimicry of Plato's style. Through the iterative process, one brings back again and again the repudiated meaning: through subservience, insubordination; and through resemblance, deception.

1.4 Iteration in python

1

As with gender theory, iteration is a core concept in programming. Iteration means repeating the same action to each bit of data in a collection of data.

2

To prepare a text for analysis, one uses iteration to process the text, one word at a time. Typically this involves "cleaning" textual elements that are undesirable for the purpose of text analysis, like like articles and prepositions whose high frequencies would obstruct word counts.⁶ Using a

⁶Elements that are highly frequent, like articles, prepositions, and even pronouns, can drown out other terms which are considered more substantial, like nouns and verbs. Similarly, irregular cases are standardized, because capital letters would count toward different words (for example, "talk" and "Talk" would be considered two different words).

loop, a program *iterates* through each word in a text, checks whether it needs to be "cleaned," and then perform a set of cleaning actions.

```
clean = []
for word in text:
    if word.isalpha():
        clean.append(word.lower())
```

The first line of code here creates an empty placeholder, `clean`, where words will be saved in their clean form. Then, the next line begins the `for` loop, which specifies a series of actions to execute on each word in the text, in this case, in Woolf's novel, *Orlando*. The loop will take the first word, "He," and then the second word, which is in this case a punctuation, an em dash, "—," and so on. The third line contains a filter, `isalpha()`, which checks if the word is comprised of only alphabetic characters (i.e. it is not a number or punctuation). If the word is alphabetic, it passes the filter (at this point, the em dash would fail this requirement, and wouldn't pass to the next line). At the next line, the word is lowercased, `lower()` and finally added to the empty list, `clean`.

The loop runs over each word in *Orlando*, one by one. When the loop is done running, the final list of words in `clean` will contain alphabetic and lowercased letters.

Below is the first sentence of the novel with all words in are lowercase and no punctuation.

```
['he', 'for', 'there', 'could', 'be', 'no', 'doubt', 'of', 'his', 'sex',
 'though', 'the', 'fashion', 'of', 'the', 'time', 'did', 'something',
 'to', 'disguise', 'it']
```

3

The next step in cleaning is to remove overly frequent words called "stop words." These words consist of prepositions, articles, pronouns, and auxiliary verbs—all of which are deemed to be semantically insubstantial relative to their frequency.

```
stop-words = ["i", "me", "my", "myself", "we", "our", "ours",
               "ourselves", "you", "your", "yours", "yourself",
               "yourselves", "he", "him", "his", "himself", "she",
```

For similar reasons, word endings and inflections like plural forms (endings in "s") and tenses ('walked', 'walks' or 'walking') are converted into a root form.

```

"her", "hers", "herself", "it", "its", "itself", "they",
"them", "their", "theirs", "themselves", "what",
"which", "who", "whom", "this", "that", "these",
"those", "am", "is", "are", "was", "were", "be", "been",
"being", "have", "has", "had", "having", "do", "does",
"did", "doing", "a", "an", "the", "and", "but", "if",
"or", "because", "as", "until", "while", "of", "at",
"by", "for", "with", "about", "against", "between",
"into", "through", "during", "before", "after", "above",
"below", "to", "from", "up", "down", "in", "out", "on",
"off", "over", "under", "again", "further", "then",
"once", "here", "there", "when", "where", "why", "how",
"all", "any", "both", "each", "few", "more", "most",
"other", "some", "such", "no", "nor", "not", "only",
"own", "same", "so", "than", "too", "very", "s", "t",
"can", "will", "just", "don", "should", "now"]

```

This step can be added as an additional filter to the above loop. After passing the filter that checks for alphabetic text, the loop will then check for stop words. If the word is *not* a stop word, it will pass to the next line to be lowercased and added to the list `clean`.

```

clean = []
for word in text:
    if word.isalpha():
        if word not in stop-words:
            clean.append(word.lower())

```

4

The following words now remain from the first sentence of Orlando:

```

['could', 'doubt', 'sex', 'though', 'fashion', 'time', 'something',
'disguise']

```

5

The final step of cleaning is to get each word's root form, the "lemma," so as to avoid counting the same word as instances of different words, like "run" and "running," or "child" and "children."

To do so, we add a line to our loop that look up each word, one by one, and returns the shortest lemma found.⁷ In the python code, the lemmatizer function `lemmatize()` is called from the `WordNetLemmatizer` module.

```
clean = []
for word in text:
    if word.isalpha():
        if word not in stop-words:
            lemma = WordNetLemmatizer.lemmatize(word)
            clean.append(lemma.lower())
```

The above code iterates through the text, passes it through filters for punctuation/numbers and stop words. If the word passes those filters, it then feeds it into the `lemmatize()` function, a kind of dictionary that looks up the word's root form. The root is saved to the variable `lemma`, which is finally lowercased and appended to `clean`. After iterating through every word in the no-stops list, and lemmatizing each one by one, the text is ready for analysis.

6

Text analysis is, at a most basic level, about counting words. From counting how frequently a word appears, one can start to derive patterns about word usage. One of the most popular text analysis libraries in python is the Natural Language Toolkit, or NLTK, written specifically for linguistic analysis. NLTK comes packaged with useful functions for simple text analysis tasks, like `concordance()`, which displays the "context window" surrounding a given word, and `similar()`, which finds words that are used in "similar" contexts to a given word.

For example, running `concordance()` on the word "woman" will output an instance of the word and its context window, a short excerpt surrounding the target word, from *Orlando*:

```
ty manly charm quality old woman loved failed growing old w
en without knowing perhaps woman heart intricate ignorance
indow pulled among cushion woman laid worn old made bury fa
red dyed cheek scarlet old woman loved queen knew man saw o
finger swore vilely master woman scarcely le bold speech le
```

⁷It uses the WordNet database, an electronic lexical database used for computational linguistics. WordNet contains every word's root form, as well as "synsets" or synonym sets, representing underlying relationships between word.

Christiane Fellbaum (1998, ed.) WordNet: An Electronic Lexical Database. Cambridge, MA: MIT Press.

ment panic twelve poor old woman parish today drink tea ton
perverse cruel disposition woman broke engagement night eve
verladen apple old bumboat woman carrying fruit market surr
embassy figure whether boy woman loose tunic trouser russia
ed stared boy ala boy must woman could skate speed vigour s
me standstill handsbreadth woman orlando stared trembled tu
d asked tumult emotion old woman answered skin bone trulls
and longer full prying old woman said stared one face bumpt
save sea bird old country woman hacking ice vain attempt d
ce melt heat pity poor old woman natural mean thawing must

11

The task is then to count these remains to find patterns. To demonstrate this process, another NLTK function, called `similar()` displays words that are used in similar contexts of the target word. This function takes a canonical concept from computational linguistics, called "distributional similarity," which measures how often terms appear in samples of language data, and assumes that words which appear in similar contexts tend to have similar meanings.

To compute the results of `similar()`, the program first takes the context of the target word from `concordance()`. Then, it counts each of the words surrounding woman. It then looks for words that have the most shared contexts, that is, the most shared words within their context window. Finally, it outputs a list of terms that share the most contexts with the word. For the word "woman", those words are the following:

```
['man', 'moment', 'night', 'boy', 'word', 'world', 'child' 'pen',  
'ship', 'door', 'one', 'room', 'window', 'light', 'little', 'lady',  
'table', 'book', 'queen', 'king']
```

12

In this sense, word meaning is understood as related to word similarity, which is computed by counting words and their neighbors.⁸

13

In fact, the simple act of counting words is also how *deep learning* methods work. Deep learning methods, like those that drive Large Language Model's text generation processes (discussed in detail in chapter 5), use word

⁸Importantly, the computer does not understand or impute *meaning* to the words. Rather, it only counts each word as a string, that is, as a piece of data composed of alphanumeric sequences.

patterns from counts to making predictions about which words ought to appear together.

In deep learning, information about word context is stored in the form of "word embeddings" or "word vectors." These embeddings or vectors contain a word's "definition," so to speak, in a machine-interpretable format. A vector for a single word, "woman," like the one below, will contain a list of numbers that represents its relationship to other words from the dataset.

Token	d1	d2	d3	...	d25
'woman'	-0.92527	-0.33879	-0.32138	...	-0.45112

The vector which represents "woman" contains a list of numbers (25 numbers across 25 dimensions) that score the similarity of the word "woman" to other words in the dataset.⁹

In reality, word vectors are organized into a matrix, into a tabular format (see below). These matrices can be relatively small, like the GLoVe matrix below, which is only 25 dimensions (the number of values per vector), or they may range to millions, billions, and most recently, trillions of vectors (the estimate of most recent models from OpenAI and Anthropic, for example). These massive vectors are the reason why Language Models is prepended with the word as "Large."

Token	d1	d2	d3	...	d25
'worst'	0.41955	0.65852	0.12606	...	-0.29946
'hoes'	0.52378	1.9645	0.051545	...	0.31016
'woman'	-0.92527	-0.33879	-0.32138	...	-0.45112
'sms'	2.1047	1.1953	-0.11219	...	-0.99524
'cuz'	-0.026389	1.3791	0.57634	...	0.10626
'gold'	-0.48171	-0.55272	-0.31395	...	-0.12601
'luck'	-0.28332	1.1739	-0.10735	...	-0.53487
'business'	0.077507	0.80513	-1.1954	...	-0.45832
'wear'	-0.41459	-0.66893	0.072897	...	-0.76915

Using this matrix format, further mathematical operations are possible with statistics, linear algebra, and calculus. For example, cosine similarity calculates the graphical distance between two vectors, that is, two different

⁹Known as the "GLoVe" dataset: Jeffrey Pennington, Richard Socher, and Christopher D. Manning. 2014. GloVe: Global Vectors for Word Representation.

words. And here is the reason they are called "vectors" in the first place—these long lists of numbers actually describe coordinates for a point on a graph. Words are thus mapped into graphical space. Word vectors thus translate relationships of word context into geometric space, giving words a numerical "meaning," so to speak.

For example, if plotting all of the words in the GLoVe dataset onto a single map, the points closest to "woman" would be the following:

```
['child', 0.9371739625930786,
'mother', 0.9214696884155273,
'whose', 0.9174973368644714,
'called', 0.9146499633789062,
'person', 0.9135538339614868,
'wife', 0.9088311195373535,
'being', 0.9037441611289978,
'father', 0.9028053283691406,
'guy', 0.9026350975036621,
'known', 0.8997253179550171,
...]
```

Here, the vector for the word "woman" is closest to that for the word "child," with a similarity score of .93, or 93%. "Woman" shares 93% of its vectors with those of "child."

Significantly, vectors make it possible to do math with language meaning. Indeed, the inventors of word vectors introduced their power to the world via a formula, one that calculates the vector of the word "Queen" (Mikolov et al. 2013). Here, the vector for the word "queen" is calculated by subtracting the vectors of "man" from "king," and finally, adding the vector for "woman."

$$\text{vector("Queen")} = \text{vector("King")} - \text{vector("Man")} + \text{vector("Woman")}$$

Like Butler's matter or *matrix*, here is another matrix that produces meaning. And this matrix similarly operates according to masculinist parameters. Gender is imbricated with conceptions of power ("King"). Furthermore, gender becomes an operand for computing language meaning. Much like Ted Underwood's graph which only understands gender when it is placed into an oppositional relationship, gender here is the *model* which shows the algorithm to be true.

16

The goal, for the following text analysis, will not be to re-draw the matrix. Rather, it will be, in Butler's words, "to show that it is occupiable" (11). In

the next section, I use iteration as a methodology that re-works text analysis procedures into a practice that deconstructs the terms of the gender binary. The goal will not be to dispel or revise the terms of the binary, but to show that these terms are *already* inhabited by that which they seek to exclude.

1.5 Fluidity

1

Subtitled *A Biography*, the novel *Orlando* takes real life as its source material. First, English literary history is reproduced and satirized, with each chapter of the novel taking a different period, like the Elizabethan, Restoration, and Victorian periods, and featuring historical figures like Queen Elizabeth and Alexander Pope as characters.¹⁰ Orlando's character herself is modeled on Woolf's friend and lover, Vita Sackville-West. Episodes from Sackville-West's life, like her affair with the author Violet Trefusis and her disinheritance of the family estate at Knole, are represented in Orlando's heartbreak with Sasha and her legal battles over her property and titles.¹¹

My text analysis method intersperses quantitative analysis with close reading, dipping in and out of the text to examine words in their original context. In doing so, it iterates through various gender terms, starting with the terms from the gender binary, "man" and "woman."

This toggling between close and distant views of language in the novel recalls a procedure that Andrew Piper calls "bifocal" reading, which "explicitly constructs the context through which something is seen as significant" (17).¹² Unlike Piper, however, who uses the quantitative component to "cede" "some measure of subjectivity" to the critical process, this method explicitly embraces the critic's inclination as the driving force of analysis (17). It is closer, in reality, to approaches described by both Tanya Clement and Katherine Bode, who view the act of computation as necessarily guided by the critic's own hand.¹³ Bode, who draws from feminist physicist Karen Barad's concept

¹⁰According to Christy L. Burns and Jane De Gay, Woolf re-works this material into parody and satire. See Burns, Christy L. "Re-Dressing Feminist Identities: Tensions between Essential and Constructed Selves in Virginia Woolf's *Orlando*." **Twentieth Century Literature**, vol. 40, no. 3, 1994, pp. 342–364. **JSTOR**, www.jstor.org/stable/441560; De Gay, Jane. "Virginia Woolf's feminist historiography in *Orlando*." *Critical Survey*, vol. 19, no. 1, 2007, p. 62+.

¹¹"If I Saw You Would You Kiss Me?": Sapphism and the Subversiveness of Virginia Woolf's *Orlando*." *PMLA*, vol. 103, no. 1, 1988, pp. 24–34. www.jstor.org/stable/462459

¹²Andrew Piper, *Enumerations*, University of Chicago Press, 2019. page 18.

¹³Karen Barad, in fact, takes up the concept of performativity which she, like Butler, traces through Derrida and JL Austin. Barad, however, describes their application as "a

of "intra-action," where objects and observers come into being through acts of observation, suggests that "we can understand our databases or models as part of the ongoing materialisation of literary texts as *emerging events* always arising from an altering how the literary past as reconfigured" (emphasis mine).¹⁴ Clement, whose multimodal work blends textual, audio, and visual registers, describes the critic as an "observer codependent" who "plays" a the work in order to open up interpretive possibilities: "One 'reads' a visualization, but to 'play' the visualisation is to engage the spatialized interpretation of that visualisation as an embodied reader in a situated context within a specific hermeneutical framework."¹⁵

In the text analysis event that follows, my personal inclination, as well as curiosity, will be essential to divert the expected path of gender's ostensible reproducibility.

3

I begin with two gender terms from the novel, "woman" and "man," represented as vectors, or points in graphical space. I then use a function, `most_similar()`, that applies cosine similarity to calculate the words "nearest" to the target words.¹⁶

```
model.wv.most_similar(positive = "woman", negative = "man")
```

```
model.wv.most_similar(positive = "man", negative = "woman")
```

In the above code expression, I added parameters specifying the most "positive" and "negative" associations between the genders. The results,

diffractive elaboration of Butler's... critical insights" (Barad, *Posthumanist Performativity*, *Signs* 2003, page 808). In contrast to Butler's focus on citation, Barad emphasizes relation and intra-action. See Barad, "Nature's Queer Performativity," 2012, footnote 27 on page 49.

¹⁴Bode, Katherine. "Computational Modeling: From Data Representation to Performative Materiality." ATNU/IES Virtual Speaker Series 2020/2021. Newcastle University, UK. November 26, 2020.

¹⁵Clement, T. (2013). Distant Listening or Playing Visualisations Pleasantly with the Eyes and Ears. *Digital Studies/le Champ Numérique*, 3(2). DOI: <http://doi.org/10.16995/dscn.236>

¹⁶The documentation describes `similar_words()` "computes cosine similarity between a simple mean of the projection weight vectors of the given keys and the vectors for each key in the model... correspond[ing] to the word-analogy and distance scripts in the original word2vec implementation.". Link to `most_similar()` source code: https://radimrehurek.com/gensim/models/keyedvectors.html#gensim.models.keyedvectors.KeyedVectors.most_similar

then, show words that are most distinctive for each gender, most strongly correlating to "woman" while also being distant from "man" in vector space.¹⁷

The terms closest to "woman" are the following:

```
[('soft', 0.3692586421966553),  
(('named', 0.34212377667427063),  
(('sciatica', 0.3223450779914856),  
(('frilled', 0.3187992572784424),  
(('despaired', 0.31375786662101746),  
(('friend', 0.31238242983818054),  
(('delicious', 0.30853813886642456),  
(('winked', 0.30514153838157654),  
(('notion', 0.3047487139701843),  
(('seductiveness', 0.30290719866752625)]
```

And those closest to "man" are the following:

```
[('chequered', 0.4025157392024994),  
(('fact', 0.3394489586353302),  
(('denounced', 0.3346075117588043),  
(('house', 0.33423593640327454),  
(('curiosity', 0.33144116401672363),  
(('defend', 0.3284823000431061),  
(('dancing', 0.3282632827758789),  
(('marbling', 0.3184848427772522),  
(('cynosure', 0.3057470917701721),  
(('rather', 0.3024100363254547)]
```

This analysis begins with an oppositional formulation of gender, one that places male and female into a positive/negative axis—in fact, much like Underwood's classification model. However, the goal here is to bring forth what has been *repudiated* by each term in the binary. Following Butler's reading of Irigaray reading Plato, my goal is to show that what is excluded from the binary is nonetheless present in the matrix which generates that binary—here, within a literal matrix of word vectors. In what follows, then, I read into a selection of these terms, examining them in context, to seek out what constitutes their meaning and, eventually, what must be repudiated in order for them to signify.

¹⁷Link to file from this project's source code: <https://github.com/gofilipa/qdr/blob/master/lib/actions.py>

I begin with a term from the list of terms similar to woman, "delicious." This term occurs five times in the novel, and only after Orlando transitions into a woman. Three of these occurrences appear in a single passage, when Orlando is sailing from Turkey back to her native England. In this passage, the ship captain offers Orlando a slice of beef. After politely refusing, Orlando launches into a rapturous speculation on the joys of womanhood.

‘A little of the fat, Ma’m?’ he asked. ‘Let me cut you just the tiniest little slice the size of your fingernail.’ At those words a *delicious* tremor ran through her frame. Birds sang; the torrents rushed. It recalled the feeling of indescribable pleasure with which she had first seen Sasha, hundreds of years ago. Then she had pursued, now she fled. Which is the greater ecstasy? The man’s or the woman’s? And are they not perhaps the same? No, she thought, this is the most *delicious* (thanking the Captain but refusing), to refuse, and see him frown. Well, she would, if he wished it, have the very thinnest, smallest shiver in the world. This was the most *delicious* of all, to yield and see him smile. ‘For nothing,’ she thought, regaining her couch on deck, and continuing the argument, ‘is more heavenly than to resist and to yield; to yield and to resist. (emphasis mine, 114)

Delicious describes a distinctly feminine kind of pleasure. Its novelty comes from its opposition to pursuit, here posed as the form of pleasure proper to the man, and that which Orlando enjoyed when he seduced Sasha, the Russian princess. In contrast to the active pleasure of the pursuer, this more passive form of pleasure is about withholding and, eventually, yielding. Yet, despite this passivity, it is not totally submissive: it implies a kind of power, the power of refusal.

What else is contained by this word, delicious? I decide to run another similarity search with "delicious" as the target word. The top result (the word most related to "delicious" in the text) is "culpable."

This term occurs in the same scene as above. Its appearance, however, is more conspicuous than the former: it emerges in a sentence where Orlando is considering her sexuality explicitly—indeed, one of the very few references in the novel to sexuality.

And as all Orlando’s loves had been women, now, through the *culpable* laggardry of the human frame to adapt itself to convention, though she herself was a woman, it was still a woman she loved; and if the consciousness of being of the same sex had any

effect at all, it was to quicken and deepen those feelings which she had had as a man. (emphasis mine, 119)

Like "delicious" above, "culpable" also describes kind of refusal, a refusal to conform to social expectations, specifically, with an obstinate sexuality. "Culpable" here modifies the "laggardry" of the body, suggesting that Orlando's persistence in her desire for women is, similarly to the refusal of the slice of beef, rooted in a kind of passivity that is not totally passive.

If in the previous example, refusal enhances a sense of power, in this one, refusal enhances passion. Power and passion, typically masculine aspects, are thus reworked into distinctly feminine terms. If femininity is constituted by these terms, what then constitutes masculinity?

Moving to the terms associated with "man," I examine the term "chequered," which appears only once, in a visually brilliant illustration of Orlando's corporeal form. After describing his aristocratic lineage, ("His fathers had been noble since they had been at all. They came out of the northern mists wearing coronets on their heads"), the narrator paints an image of Orlando stepping into the sunlight, into "the yellow pools which *chequered* the floor" (12). This depiction luxuriates over Orlando's face and body:

Orlando stood now in the midst of the yellow body of an heraldic leopard. When he put his hand on the window-sill to push the window open, it was instantly coloured red, blue, and yellow like a butterfly's wing. Thus, those who like symbols [...] might observe that though the shapely legs, the handsome body, and the well-set shoulders were all of them decorated with various tints of heraldic light, Orlando's face, as he threw the window open, was lit solely by the sun itself. (12)

This image of Orlando's body *chequered* into different colored pieces is indeed a symbol—but one of division rather than nobility. This sense of division is exacerbated by the narration itself, whose increasing detours into extradiegetic reflection heavily layers the the narrator's rendition of events over the story, a rendition that soon gallops into full-fledged figuration:

Directly we glance at Orlando standing by the window, we must admit that he had eyes like drenched violets, so large that the water seemed to have brimmed in them and widened them; and a brow like the swelling of a marble dome pressed between the two blank medallions which were his temples. Directly we glance at eyes and forehead, thus do we rhapsodize. Directly we glance

at eyes and forehead, we have to admit a thousand disagreeables which it is the aim of every good biographer to ignore. 12-13

The narrator endows Orlando's beauty with mystical qualities—his eyes like flowers, his forehead like marble. His evocative language undermines his own pretense to objectivity, as a self-avowed "scribe" committed to straightforward description. This scene is the first in many scenes in the novel where the narrator calls into question his credibility as a biographer. These momentary slips into figurative language grow into a crisis of signification that recurs persistently through the novel.

In one of the scenes where this crisis appears most strongly contains a term from the "woman" list, "despaired." Like "chequered," this term occurs at a moment when figurative language is called directly into question. In this scene, Orlando is in the midst of a period of depression following his abandonment by Sasha. He is lying beneath an Oak tree, ruminating about the meaning of truth, love, and above all, language itself.

‘The sky is blue,’ he said, ‘the grass is green.’ Looking up, he saw that, on the contrary, the sky is like the veils which a thousand Madonnas have let fall from their hair; and the grass fleets and darkens like a flight of girls fleeing the embraces of hairy satyrs from enchanted woods. ‘Upon my word,’ he said (for he had fallen into the bad habit of speaking aloud), ‘I don’t see that one’s more true than another. Both are utterly false.’ And he despaired of being able to solve the problem of what poetry is and what truth is and fell into a deep dejection. 75

If, in the previous scene, Orlando was divided by different colors of light, here he appears divided by doubt about language and its ability to accurately describe reality. He first attempts plain language, "the sky is blue," "the grass is green." But these prove insufficient to describe a sky of "veils [from] a thousand Madonnas" and grass that "fleets and darkens" like "girls fleeing." Orlando, like the narrator, cannot make language do the work of expression. So he turns to figurative elements, comparisons of feminine subtlety and evasion, which betray his pain from Sasha's betrayal.

As a term, "despair" crosses the lines of the binary. It comes from the "feminine" list, but unlike most of the other terms on the list, it describes a moment in the narrative when Orlando is still a man. In other words, like Orlando, this term has gender-crossing capabilities.

Indeed, throughout the novel, language and gender appear to be co-constructed. An interesting passage, which contains words from both the

woman and man lists, shows how language's failure to describe reality is related to how gender expresses itself.

In this passage, Orlando beholds Sasha for the first time, and he cannot tell whether she is a man or a woman. In his tumultuous and agonizing attempt to determine her gender, the words "curiosity," from the "man" list of similar words, and "seductiveness," from the "woman" list, both appear.

He beheld, coming from the pavilion of the Muscovite Embassy, a figure, which, whether boy's or woman's, for the loose tunic and trousers of the Russian fashion served to disguise the sex, filled him with the highest *curiosity*. The person, whatever the name or sex, was about middle height, very slenderly fashioned, and dressed entirely in oyster-coloured velvet, trimmed with some unfamiliar greenish-coloured fur. But these details were obscured by the extraordinary seductiveness which issued from the whole person. Images, metaphors of the most extreme and extravagant twined and twisted in his mind. He called her a melon, a pineapple, an olive tree, an emerald, and a fox in the snow all in the space of three seconds; he did not know whether he had heard her, tasted her, seen her, or all three together. (For though we must pause not a moment in the narrative we may here hastily note that all his images at this time were simple in the extreme to match his senses and were mostly taken from things he had liked the taste of as a boy. But if his senses were simple they were at the same time extremely strong. To pause therefore and seek the reasons of things is out of the question). . . . A melon, an emerald, a fox in the snow—so he raved, so he stared. When the boy, for alas, a boy it must be—no woman could skate with such speed and vigour—swept almost on tiptoe past him, Orlando was ready to tear his hair with vexation that the person was of his own sex, and thus all embraces were out of the question. (27-28)

At the same time which Orlando cannot place Sasha's gender, he also cannot find the right words to describe her, and turns to arbitrary metaphors, "A melon, a pineapple, an olive tree, an emerald, a fox in the snow." This instability of language, and its relation to an indecipherable gender, is exacerbated by narrative veering between Orlando's thoughts and those of the narrator. In particular, the insertion of the narrator's "pause," a pause which purports to explain Orlando's arbitrary metaphors, breaches the fourth wall,

reinforcing the constructed quality of the scene. Thus, both gender and language appear to be constituted through one another. Taking words from both the "woman" and "man" lists of similar words, this passage suggests that the two genders intersect via language's unstable, expressive qualities. Gender, like language, is *fluid*.

This iterative reading practice, which moves from close to distant, brings certain terms within matrix of gender to the surface. From the graphical orbit of "man" and "woman," terms like "delicious," "culpable," "chequered," "despaired," and in the longer passage, "seductiveness" and "curiosity," show that either gender is constituted in ways that disrupt traditional expectations. Through an association with refusal, the terms similar to "woman" rework feminine passivity into power and passion. The masculine gender, by contrast, is depicted in terms of division, which spreads from Orlando's interior experience to the extradiegetic layer of the narration, and signals a larger conflict with language and storytelling.

1.6 Conclusion

The discursivity of gender, however, has a major blindspot—the body. Trans Studies scholar Jay Prosser, argues that Queer Studies in general, and Butler's theory of Gender Peformativity in particular, has a tendency to overlook the materialities of gendered embodiment in favor of discursive explorations of gender. Transness is often discussed in terms which it avoids: subversion. As Prosser points out, the concept of subversion, of resisting dominant gender norms, of boundary crossing, "cannot account for a transsexual desire for sexed embodiment as *telos*" (33).

And particularly, the novel *Orlando*, according to Prosser, bears out this critique of gendered embodiment. Here, Woolf's experimentation with language and narrative form idealizes the eponymous character's relationship to gender. Prosser says that, "Orlando is not about the sexed body at all but the cultural vicissitudes of gender... Orlando is free to move beyond h/er body—quite queerly, to break through the limits of the flesh" (168).

Not only is the reality of the sexed body elided throughout the novel, but also the reality of the racialized one. The racialized body, as some critics point out, forms a substrate for the construction of gender norms. And indeed, this dimension of race has been so far unacknowledged in this analysis. Careful readers of the novel *Orlando* will have noticed that, in the introduction of the chapter, I left out the second half of the first sentence, which appears here in italics: "He—for there could be no doubt of his sex, though the fashion of the time did something to disguise it—*was in the act*

of slicing at the head of a Moor which swung from the rafters."

The elements of imperial and racist violence in the visual of the decapitated moor are not benign, they directly affect the analysis of gender in the novel. As Jamie Hovey explains,

Beginning with an image of a decapitated Moor, Orlando appropriates the exoticized subjects of colonial discourse to create for its protagonist a sexuality that is not bourgeois or heterosexual. However, Orlando's playful but insistent production of white femininity allows the protagonist to pass as respectable and heterosexual by displacing her transgressive sexuality onto racial others, thereby masking Orlando's masculine-identified literary and sexual desires. (Hovey 398).

Like *Orlando's* conspicuous narrator, there is something conspicuous about my own reading of this story, and its elision of matters of race. I have left it out because, as I argue over the next chapters, the tool of analysis should match the constraint under examination. And Gender Performativity, while being an effective tool for analyzing the constraint of gender as an iterative form, a fluid form, it cannot be applied toward an analysis of gendered embodiment—a problem that I turn to in chapters four and five, through the lenses of Black Feminist Studies and Trans Studies.

A decade after Butler developed her concept of Gender Performativity, Omise'eke Natasha Tinsley writes about the problem of gender fluidity as a metaphor. Her critique re-works the trope of fluidity which, drawing from the ocean, "is not an easy metaphor or queer and racially hybrid identities but for concrete, painful, and liberatory experience" (192-193). For Tinsley, fluidity is an opportunity for "a return to the materiality of water to make its metaphors mean more complexly, shaking off settling into frozen figures" (212). Tinsley theorizes fluidity as being stripped by the water, particulars of identity washed away in the current. She explains that "brown bodies are gender fluid not because they choose parodic proliferations but because they have been 'washed of all this lading, bag and baggage'" (209).